Semester project

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RQ:

Does a high carbon tax rate lead to lower GHG emissions per capita after factoring in fossil fuel share in the energy mix and GDP?

```
library(tidyverse)
library(readxl)

Carbon_tax <- read_excel("Carbon_tax.xlsx", sheet = "Compliance_Price", skip = 1)
GHG <- read_csv("GHG.csv")
Population <- read_csv("Population.csv")
GDP <- read_csv("GDP.csv")
Energy_mix <- read_csv("Energy_mix.csv")
```

(Tax rate is in Euros per tonne of CO2-equivalent)

```
#Choosing and renaming relevant columns
My_countries <- c("Austria", "Belgium", "Denmark", "Estonia", "Finland",
                  "France", "Germany", "Greece", "Hungary", "Iceland",
                  "Ireland", "Italy", "Latvia", "Lithuania", "Luxembourg",
                  "Netherlands", "Norway", "Poland", "Portugal",
                  "Slovenia", "Spain", "Sweden")

Year_range <- 2005:2024

Carbon_tax_clean <- Carbon_tax %>%
  filter(
    `Instrument Type` == "Carbon tax"
    ) %>% #filtering out ETS only keeping carbon tax
  mutate(
    Country = str_extract(`Name of the initiative`,
```

```
  paste(My_countries, collapse = "|"))
  ) %>%
  filter(!is.na(Country)) %>%
  select(
    Country, all_of(as.character(Year_range))
    ) %>%
  pivot_longer(
    cols = -Country, names_to = "Year", values_to = "Tax_rate_usd"
    ) %>%
  mutate(
    Year = as.numeric(Year),
    Tax_rate_usd = ifelse(is.na(Tax_rate_usd), 0, Tax_rate_usd)
    )

no_tax_countries <- setdiff(My_countries, unique(Carbon_tax_clean$Country))

no_tax_filled <- expand_grid(Country = no_tax_countries, Year = Year_range
    ) %>%
  mutate(Tax_rate_usd = 0)

Carbon_tax_clean <- bind_rows(Carbon_tax_clean, no_tax_filled) %>%
  arrange(Country, Year)

#Choosing and renaming relevant columns
GHG_clean <- GHG %>%
  select(
    `Geopolitical entity (reporting)`, TIME_PERIOD, OBS_VALUE
    ) %>%
  rename(
    Country = `Geopolitical entity (reporting)`,
    Year = TIME_PERIOD,
    Total_emissions = OBS_VALUE
  )

#Choosing and renaming relevant columns
Population_clean <- Population %>%
  select(
    `Geopolitical entity (reporting)`, TIME_PERIOD, OBS_VALUE
  ) %>%
  rename(Country = `Geopolitical entity (reporting)`,
         Year = TIME_PERIOD,
         Population = OBS_VALUE
```

```
        )
#Choosing and renaming relevant columns
GDP_clean <- GDP %>%
  select(
    `Geopolitical entity (reporting)`, TIME_PERIOD, OBS_VALUE
  ) %>%
  rename(
    Country = `Geopolitical entity (reporting)`,
    Year = TIME_PERIOD,
    GDP_per_capita = OBS_VALUE
  )

#Choosing and renaming relevant columns
Energy_mix_clean <- Energy_mix %>%
  select(
    Country = `Geopolitical entity (reporting)`,
    Year = TIME_PERIOD,
    Source = `Standard international energy product classification (SIEC)`,
    OBS_VALUE
  ) %>%
  filter(Source %in% c("Solid fossil fuels",
  "Natural gas",
  "Oil and petroleum products (excluding biofuel portion)", "Total")
        ) %>%
  pivot_wider(
    names_from = Source, values_from = OBS_VALUE, values_fill = 0
    ) %>%
  mutate(
    Fossil_fuels_share = (`Solid fossil fuels` + `Natural gas` +
    `Oil and petroleum products (excluding biofuel portion)`)
    / Total * 100
    ) %>%
  select(
    Country, Year, Fossil_fuels_share
    )
```

```
Merged1 <- merge(
  Carbon_tax_clean, GHG_clean,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )

Merged2 <- merge(
```

```
  Population_clean, GDP_clean,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )

Merged3 <- merge(
  Merged1, Merged2,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )

Trial_dataset <- merge(
  Merged3, Energy_mix_clean,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )
```

Calculating: emissions per capita = Total emissions/ Total population

Unit of measure is in tonnes

```
Trial_dataset$Emissions_per_capita <-
  Trial_dataset$Total_emissions * 1000000 / Trial_dataset$Population

Final_dataset <- Trial_dataset %>%
  select(
    Country, Year, Tax_rate_usd, GDP_per_capita, Fossil_fuels_share,
    Emissions_per_capita
  )
```